

Research Notes on K-means clustering

K-means clustering

$$J = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2$$

>> Section 1

Proper initialization is very important when it comes to finding global optimum in a clustering algorithm. Random initialization is often used for k means clustering, but this can result in poor cluster quality. This is why the k-means++ initialization algorithm was developed. The algorithm attempts to find initial centers that are as close as possible to the global optimum, reducing computing time and producing more accurate clusters. The k-means++ algorithm works by spreading out the initial cluster centers. It selects the first center uniformly at random from the dataset. Each subsequent center is chosen stochastically, with the probability of a point being selected proportional to its contribution to the overall error. This method achieves an $O(\log k)$ approximation of the optimum solution.

>> Section 2

Pseudocode The below pseudocode outlines the implementation of the standard k-means clustering algorithm. Initialization of centroids, distance metric between points and centroids, and the calculation of new centroids are design choices and will vary with different implementations. In this example pseudocode, distance() returns the distance between the specified points.

>> Section 3

Assignment step: Assign each observation to the cluster with the nearest mean (centroid): that with the least squared Euclidean distance. (Mathematically, this means partitioning the observations according to the Voronoi diagram generated by the means.) $S_i(t) = \{x_p : \|x_p - \mu_i(t)\|^2 \leq \|x_p - \mu_j(t)\|^2 \forall j, 1 \leq j \leq k\}$,

$S_{i^{(t)}} = \{x_p : \|x_p - \mu_{i^{(t)}}\|^2 \leq \|x_p - \mu_{j^{(t)}}\|^2 \forall j, 1 \leq j \leq k\}$, where each x_p is assigned to exactly one $S_i(t)$, even if it could be assigned to two or more of them. **Update step:** Recalculate means (centroids) for observations assigned to each cluster. This is also called refitting. $\mu_i(t+1) = \frac{1}{|S_i(t)|} \sum_{x_j \in S_i(t)} x_j$

>> Section 4

Finding the optimal number of clusters (k) for k-means clustering is a crucial step to ensure that the clustering results are meaningful and useful. Several techniques are

available to determine a suitable number of clusters. Here are some of commonly used methods:

>> Section 5

In the worst-case, Lloyd's algorithm needs $i = 2^{\Omega(\sqrt{n})}$ iterations, so that the worst-case complexity of Lloyd's algorithm is superpolynomial. Lloyd's k-means algorithm has polynomial smoothed running time. It is shown that for arbitrary set of n points in $[0, 1]^d$, if each point is independently perturbed by a normal distribution with mean 0 and variance σ^2 , then the expected running time of k-means algorithm is bounded by $O(n^{34} k^{34} d^8 \log^4(n/\sigma^6))$, which is a polynomial in n , k , d and $1/\sigma$. Better bounds are proven for simple cases. For example, it is shown that the running time of k-means algorithm is bounded by $O(d n^4 M^2)$ for n points in an integer lattice $\{1, \dots, M\}^d$. Lloyd's algorithm is the standard approach for this problem. However, it spends a lot of processing time computing the distances between each of the k cluster centers and the n data points. Since points usually stay in the same clusters after a few iterations, much of this work is unnecessary, making the naive implementation very inefficient. Some implementations use caching and the triangle inequality in order to create bounds and accelerate Lloyd's algorithm.

>> Section 6

k-means clustering is a method of vector quantization, originally from signal processing, that aims to partition n observations into k clusters in which each observation belongs to the cluster with the nearest mean (cluster centers or cluster centroid). This results in a partitioning of the data space into Voronoi cells. k-means clustering minimizes within-cluster variances (squared Euclidean distances), but not regular Euclidean distances, which would be the more difficult Weber problem: the mean optimizes squared errors, whereas only the geometric median minimizes Euclidean distances. For instance, better Euclidean solutions can be found using k-medians and k-medoids. The problem is computationally difficult (NP-hard); however, efficient heuristic algorithms converge quickly to a local optimum. These are usually similar to the expectation-maximization algorithm for mixtures of Gaussian distributions via an iterative refinement approach employed by both k-means and Gaussian mixture modeling. They both use cluster centers to model the data; however, k-means clustering tends to find clusters of comparable spatial extent, while the Gaussian mixture

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model allows clusters to have different shapes. The unsupervised k-means algorithm has a loose relationship to the k-nearest neighbor classifier, a popular supervised machine learning technique for classification that is often confused with k-means due to the name. Applying the 1-nearest neighbor classifier to the cluster centers obtained by k-means classifies new data into the existing clusters. This is known as nearest centroid classifier or Rocchio algorithm.

>> Section 7

Early applications of the k-means algorithm were primarily found in signal processing and data compression, particularly in the context of vector quantization. Lloyd's work at Bell Labs focused on representing analog signals using a limited set of discrete values, where clustering was used to reduce the amount of data required while preserving signal quality. As computing power increased, k-means clustering became widely used in pattern recognition and statistical classification due to its simplicity and computational efficiency. It was later adopted in early machine learning and data analysis tasks involving large datasets. Despite its widespread use, limitations such as sensitivity to initial centroid placement and difficulty handling non-spherical clusters were recognized early on, motivating the development of improved clustering methods and initialization techniques. Numerous extensions of k-means have since been developed to address limitations of the original algorithm, including methods such as fuzzy c-means, which allows data points to belong to multiple clusters with varying degrees of membership, and kernel k-means, which uses kernel functions to identify non-linearly separable clusters.